

**To the Editor-in-Chief,**  
***Knowledge-Based Systems***

Dear Editor,

We are pleased to submit our original research article entitled "**PDT: Predictive Domain Transform for Multivariate Time Series Forecasting**" for your consideration for publication in *Knowledge-Based Systems*.

Multivariate Time-Series Forecasting (MTSF) is a cornerstone of decision-making in modern industrial intelligence systems. While recent linear-based models have challenged the dominance of Transformers due to their efficiency, they inherently struggle to represent complex non-linear temporal patterns. Conversely, deep neural networks often incur high computational costs and risk overfitting. Addressing this critical trade-off is essential for the advancement of robust knowledge-based forecasting systems.

In this work, we propose a novel framework named **PDT**, which effectively bridges the gap between linear efficiency and non-linear representation capacity. Our contributions are threefold and align closely with the scope of your journal:

1. **Methodological Innovation:** We introduce a **Predictive Domain Transform Method (PDTM)**. Unlike traditional fixed-basis transforms (e.g., Fourier or Wavelet) used in previous works, PDTM is data-adaptive and explicitly optimized to minimize prediction error. It projects time series into a "prediction-optimal" latent space, exhibiting a strong energy concentration property that facilitates efficient feature extraction.
2. **Architectural Novelty:** We design a **Dual-Route architecture** that processes the series in parallel: a Linear-Route efficiently extrapolates dominant linear trends, while an Encoder-Route captures complex non-linear dynamics.
3. **Advanced Dependency Modeling:** To resolve the conflict between Channel-Independent (CI) and Channel-Dependent (CD) paradigms, we propose a **Masked Channel Dependent (MCD)** strategy. This mechanism utilizes a learnable mask to selectively fuse information only from strongly correlated variates, effectively suppressing noise in high-dimensional data.

Our extensive experiments on eight real-world datasets (including Traffic, Electricity, and ETT) demonstrate that PDT achieves state-of-the-art performance, surpassing recent baselines such as iTransformer, PatchTST, and TimeMixer. Notably, our model achieves a superior balance between accuracy and computational efficiency, making it highly suitable for practical deployment in intelligent systems.

We believe that this study makes a significant contribution to the field of time-series data mining and intelligent systems, which fits well within the aims and scope of *Knowledge-Based Systems*.

We confirm that this manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the manuscript and agree with its submission to *Knowledge-Based Systems*.

Thank you for your time and consideration.

Sincerely,

A handwritten signature in black ink, appearing to be 'Jiangang Lu'.

**Jiangang Lu** (Corresponding Author)

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